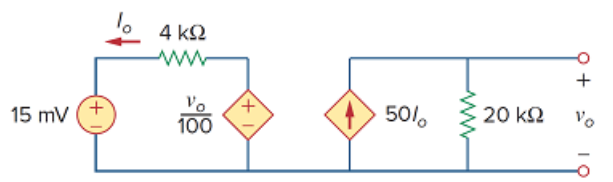


Problem

Calculate v_o and I_o in the circuit of Fig. 3.121.

Figure 3.121:



Step-by-step solution

Step 1 of 4

Draw the circuit diagram as shown in Figure 1.

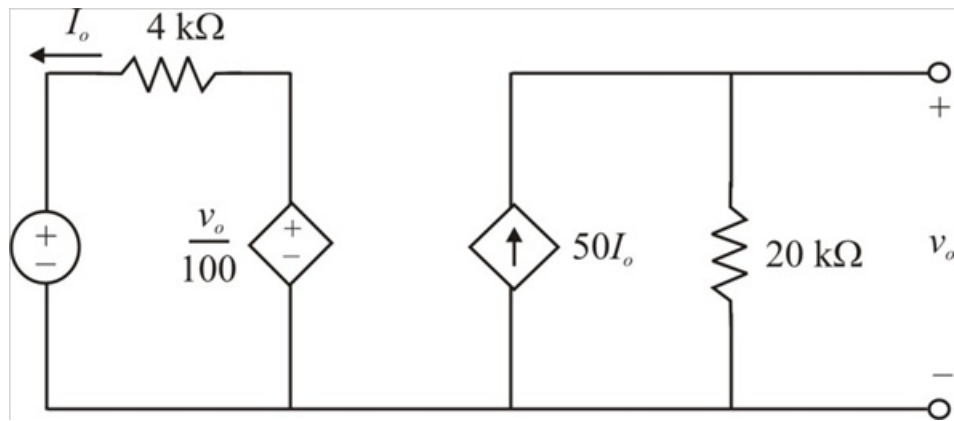


Figure 1

Step 2 of 4

The circuit is labeled and the mesh Loops are shown in Figure 2.

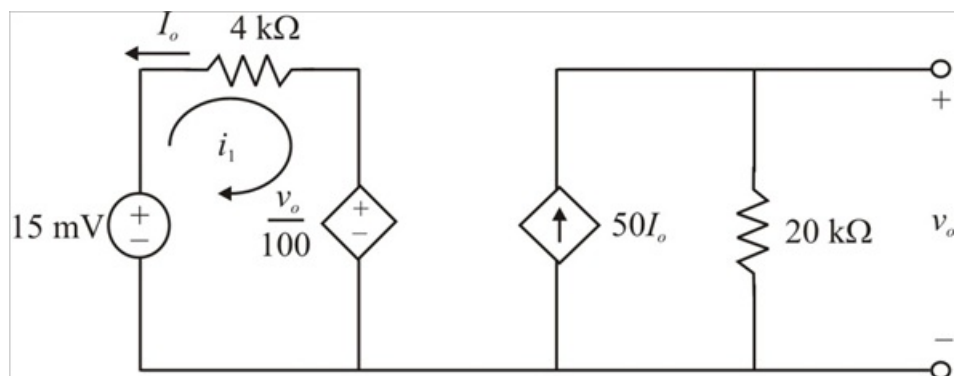


Figure 2

Step 3 of 4

For the input Loop 1, Kirchhoff's voltage law gives,

$$-(15 \times 10^{-3}) + (4 \times 10^3)i_1 + \frac{v_o}{100} = 0 \dots\dots (1)$$

But, $i_1 = -I_o$

The current through 20 k Ω resistor is equal to $50I_o$. The voltage v_o is equal to the voltage across the 20 k Ω resistor. Applying Ohm's law,

$$\begin{aligned} v_o &= iR \\ &= (50I_o)(20 \times 10^3) \\ &= 1000 \times 10^3 I_o \end{aligned}$$

Substituting $i_1 = -I_o$ and v_o in equation (1), we get

$$\begin{aligned} -(15 \times 10^{-3}) - (4 \times 10^3)I_o + \frac{1000 \times 10^3 I_o}{100} &= 0 \\ (6 \times 10^3)I_o &= 15 \times 10^{-3} \\ I_o &= \frac{15 \times 10^{-3}}{6 \times 10^3} \\ &= 2.5 \mu\text{A} \end{aligned}$$

Therefore the current I_o is $2.5 \mu\text{A}$.

Step 4 of 4

The voltage v_o can be calculated as:

$$\begin{aligned} v_o &= 1000 \times 10^3 I_o \\ &= 1000 \times 10^3 (2.5 \times 10^{-6}) \\ &= 2.5 \text{ V} \end{aligned}$$

Therefore the voltage v_o is 2.5 V .